

# Advanced Deployment Gates Configuration

## Pre-Deployment Gates

### 1. Query Work Items Gate

**Purpose:** Ensure no critical bugs before deployment

- Gate Configuration:
- Query: Work Item Type = Bug AND Priority = 1 AND State <> Closed
  - Success criteria: Query returns 0 items
  - Timeout: 60 minutes
  - Sampling interval: 5 minutes

**Business Rule:** No P1 bugs allowed in production deployment

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### 2. Azure Monitor Gate

**Purpose:** Check application health metrics

- Alert Rule Query:
- ```
{
  "query": "requests | where success == false | summarize failureRate = (count() * 100.0) / toint(count()) | where failureRate > 5",
  "timeRange": "PT15M"
}
```
- Gate Configuration:
- Success criteria: No active alerts
  - Timeout: 30 minutes
  - Sampling interval: 5 minutes

**Business Rule:** Error rate must be below 5% before deploying

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### 3. Security Scan Gate

**Purpose:** Ensure no high-severity vulnerabilities

- Gate Configuration:
- Integration: Azure Security Center API
  - Check: No high/critical vulnerabilities in last 24 hours
  - Success criteria: Compliance score > 90%
  - Timeout: 15 minutes

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## 4. ServiceNow Change Management

**Purpose:** Compliance with ITIL change management

- Gate Configuration:
- API endpoint: ServiceNow REST API
  - Query: Change ticket status for release
  - Success criteria: Change Request status = "Approved"
  - Required fields:
    - Change Number
    - Approver
    - Implementation Window

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## Post-Deployment Gates

### 1. Health Monitoring Gate

**Purpose:** Verify deployment success through metrics

- Metrics to Monitor:
- └─ HTTP 5xx errors: < 1% of requests
  - └─ Response time P95: < 500ms
  - └─ CPU utilization: < 70%
  - └─ Memory usage: < 80%
  - └─ Active connections: Within normal range
- Gate Configuration:
- Delay before evaluation: 10 minutes (warmup period)
  - Sampling interval: 5 minutes
  - Timeout: 60 minutes
  - Success criteria: All metrics within threshold for 3 consecutive checks

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### 2. Synthetic Transaction Monitoring

**Purpose:** Validate key user flows

Synthetic Tests:

- 1. User Login → Success
- 2. Browse Products → Response < 2s
- 3. Add to Cart → Success
- 4. Checkout Process → Complete transaction
- 5. Order Confirmation → Received

Gate Configuration:

- Test frequency: Every 3 minutes
- Success criteria: 95% pass rate over 30 minutes
- Automatic rollback if < 90%

3. Business Metrics Validation

**Purpose:** Ensure deployment doesn't break business KPIs

Metrics to Track:

- └─ Conversion rate: ±5% of baseline
- └─ Average order value: ±10% of baseline
- └─ New user registrations: ≥ 90% of baseline
- └─ Search success rate: ≥ 95%

Comparison Period:

- Baseline: Last 7 days before deployment
- Evaluation: First 2 hours after deployment

Automated Rollback Configuration

Rollback Triggers

yaml

Automatic Rollback Conditions:

- Error rate > 10% for 5 consecutive minutes
- Response time P95 > 2000ms for 10 minutes
- Availability drops below 99%
- Any synthetic test failure rate > 20%
- Business metric drops > 25% from baseline

Rollback Strategy

- 1. Immediate traffic switch to previous version
- 2. Preserve new version for investigation

- 3. Notify incident response team
- 4. Create post-mortem work item
- 5. Log detailed metrics for analysis

## Notification Configuration

### Approval Notifications

- Pre-Deployment Approval Required:
- Notification to: Deployment Approvers group
  - Include information:
    - Build version
    - Changes included (git commits)
    - Test results summary
    - Security scan results
    - Gate evaluation status
  - Timeout: 12 hours
  - Auto-reject after timeout

### Deployment Status Notifications

- Channels:
- Microsoft Teams: #deployments channel
  - Email: DevOps team distribution list
  - PagerDuty: On-call engineer (if failures)
  - ServiceNow: Update change ticket

- Notification Events:
- Deployment started
  - Gates evaluation status
  - Approval pending
  - Deployment succeeded
  - Deployment failed
  - Rollback triggered

## Example: Complete Production Deployment Flow

1. Artifact Available from Build Pipeline

## 2. Pre-Deployment Gates (30 min timeout)

- └─ Check P1 bugs: PASS (0 bugs)
- └─ Azure Monitor: PASS (no alerts)
- └─ Security scan: PASS (compliance 95%)
- └─ ServiceNow ticket: PASS (approved)



## 3. Pre-Deployment Approval

Approvers: CTO + DevOps Lead  
Status: APPROVED (both approved in 2 hours)



## 4. Deployment Tasks Execute

- └─ Deploy to blue environment
- └─ Warmup application (5 min)
- └─ Run smoke tests: PASS
- └─ Switch 10% traffic to blue



## 5. Post-Deployment Gates (60 min monitoring)

Time 0-15 min: 10% traffic on new version

- └─ Error rate: 0.3% ✓
- └─ Response time: 425ms ✓
- └─ Synthetic tests: 100% pass ✓

Time 15-30 min: 50% traffic on new version

- └─ Error rate: 0.5% ✓
- └─ Response time: 450ms ✓
- └─ Business metrics: +2% conversion ✓

Time 30-60 min: 100% traffic on new version

- └─ All metrics stable ✓
- └─ Deployment SUCCESSFUL

## Best Practices

1. **Start Conservative:** Begin with stricter gates, loosen as confidence grows

2. **Monitor Everything:** If you can't measure it, you can't gate on it
3. **Fast Feedback:** Keep gate evaluation intervals short (3-5 minutes)
4. **Clear Ownership:** Every gate should have a responsible team
5. **Document Thresholds:** Why did you choose 5% error rate vs 10%?
6. **Test Gates:** Simulate failures to verify rollback works
7. **Review Regularly:** Update gates based on incident learnings